TECHNICAL EVIDENCE

RESEARCH AGENDA

Database, ML Models, And Himalayan Accuracy Path

PLAIN-ENGLISH PURPOSE

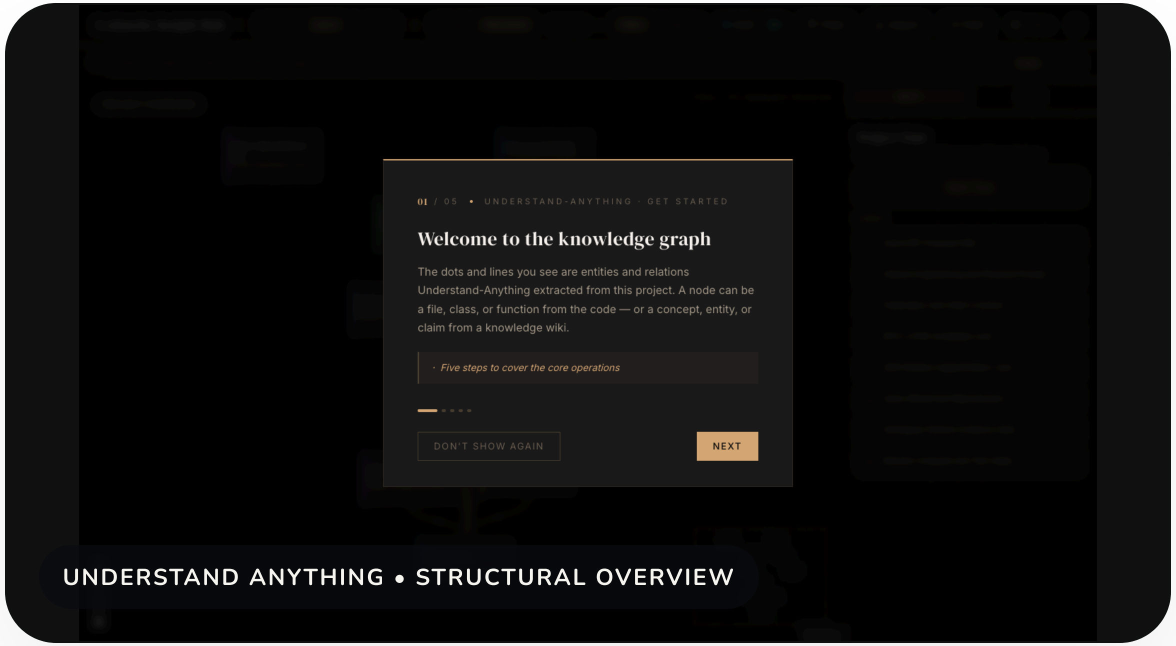
This deck explains the system as three connected parts: a database that stores facts, machine-learning models that estimate risk, and evidence gates that decide what may be shown or claimed.

For beginners

No code reading required; every technical term is mapped to a simple operational meaning.

For scientists

The deck keeps label quality, validation data, SAR limits, and local holdout gates visible.



ONE-MINUTE MENTAL MODEL

● TECHNICAL EVIDENCE

Three Things To Remember

1 · DATABASE

Stores the facts

Supabase/Postgres stores forecast runs, grid cells, events, model status, evaluation records, scientist reviews, and daily verification records.

2 · MODELS

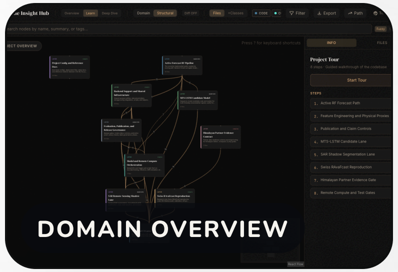
Estimate risk

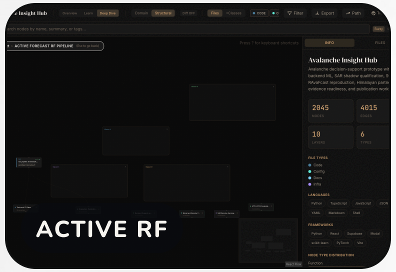
Random Forest is the active structured scorer; MTS-LSTM, SAR segmentation, and Swiss RAvaFcast remain gated research or candidate lanes.

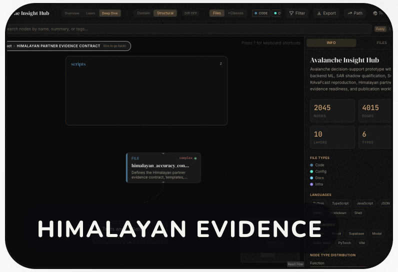
3 · EVIDENCE GATES

Control claims

Publication, SAR, Himalayan partner, and holdout gates stop unsupported claims even when a model or graph looks promising.







DASHBOARD MAP

What The Graph Is Showing

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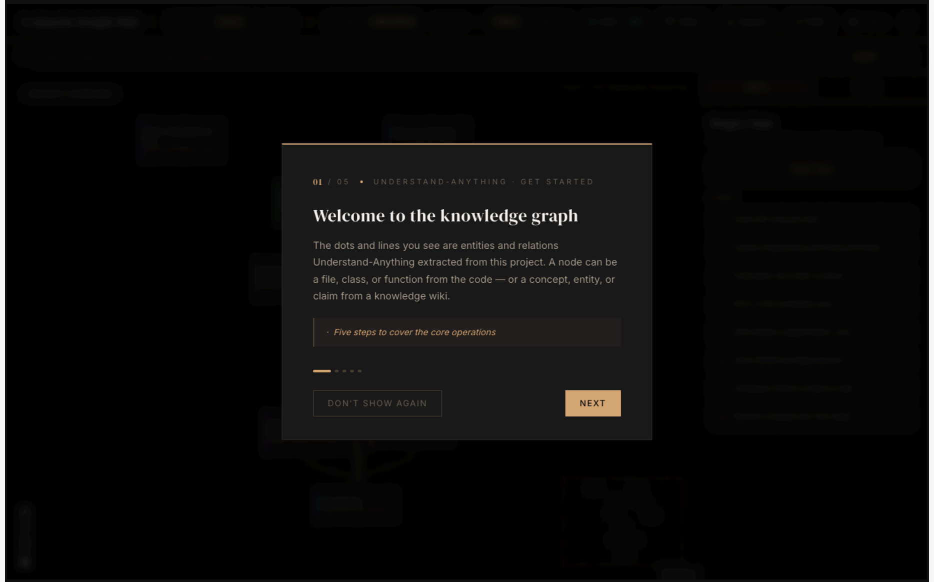

10 STRUCTURAL LAYERS

NODES

2,045

Files, functions, classes, pipelines, config, and docs.

EDGES

4,015

Relationships that show how code and evidence connect.

LAYERS

10

Grouped ML/backend workstreams, not random file lists.

The graph is a navigation aid. It does not prove scientific accuracy by itself; it helps reviewers see where the implemented evidence and remaining gates live.

DATABASE BASICS

The Database In Plain English

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FORECAST FACTS

Runs and grid cells

`forecast_runs`, `forecast_run_hours`, `forecast_grids`, and publication events record what was generated and published.

GROUND EVIDENCE

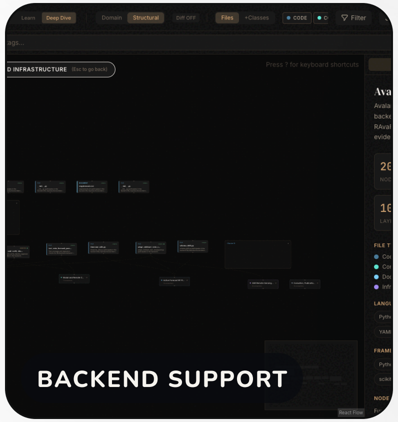
Events and reports

`avalanche_events`, field reports, outcome labels, and feature logs form the evidence trail behind model evaluation.

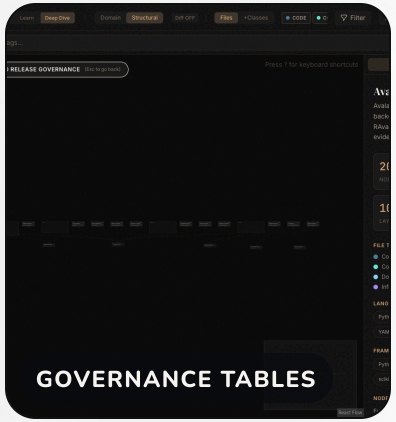
GOVERNANCE

Status and reviews

`model_status`, `evaluation_runs`, `scientist_validation_reviews`, and daily verifications keep model state and human checks visible.



BACKEND SUPPORT



GOVERNANCE TABLES

DATABASE FLOW

Forecast Run To Public Map

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CORE FLOW

INPUTS

Weather, terrain, snowpack proxies, event history, and model status.

FORECAST RUN

Backend scoring creates a dated run with lineage and metadata.

GRID CELLS

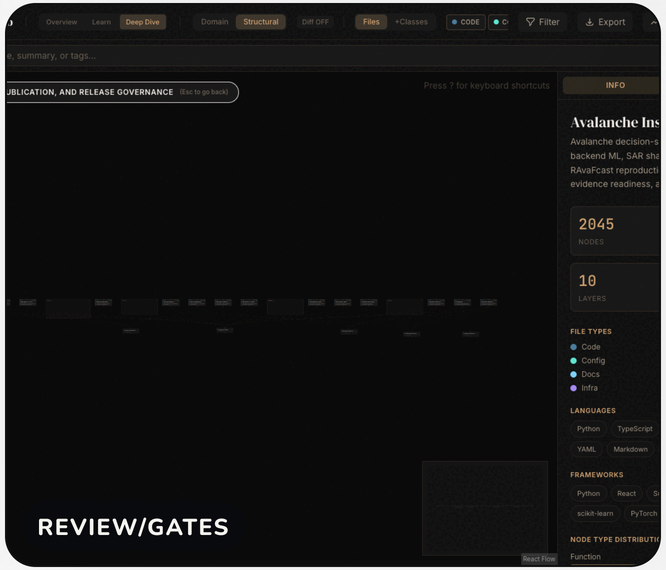
Cell-level risk values become the map surface and bulletin context.

REVIEW

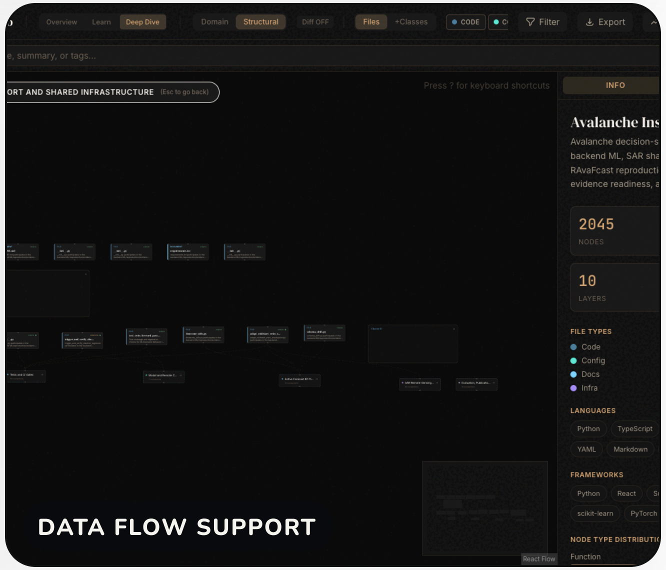
Scientists and operators inspect cases, actions, and daily verification records.

WHAT IS NOT UI-COMPLETE TODAY

Himalayan partner evidence intake is still CSV, source-manifest, and CLI based. The current UI supports product and review concepts, but the full partner evidence package is not yet a web form workflow.



REVIEW/GATES



DATA FLOW SUPPORT

MODEL LADDER

Current ML Model Map

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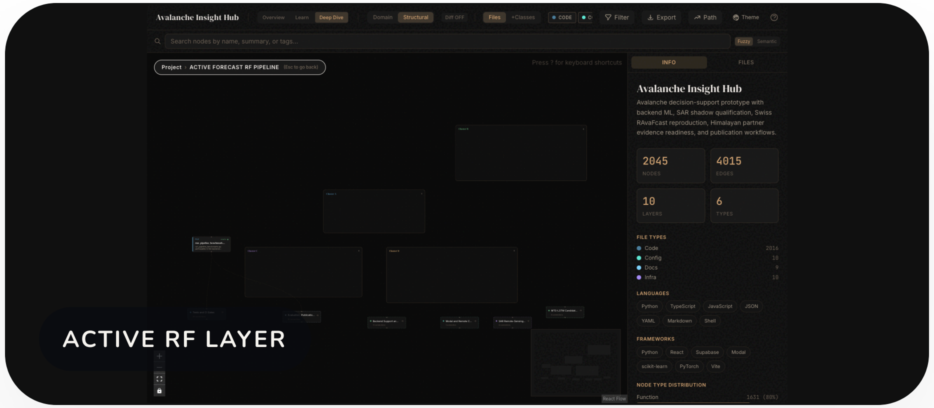


Five lanes

Active
Random Forest with terrain, weather, and snowpack features.

Candidate
MTS-LSTM for future temporal learning.

Shadow and research
SAR shadow plus Swiss RAvaFcast RF4, GPxyz, and refined aggregation.

Evidence and claim lock
Himalayan partner evidence; `production_scoring_allowed=false`; `himalayan_accuracy_claim_allowed=false`.

TECHNICAL EVIDENCE

How Random Forest Improves Prediction

INPUT FEATURES

More than one signal

Terrain, weather, snowpack proxies, historical context, and freshness cues become structured columns for the model.

TREE ENSEMBLE

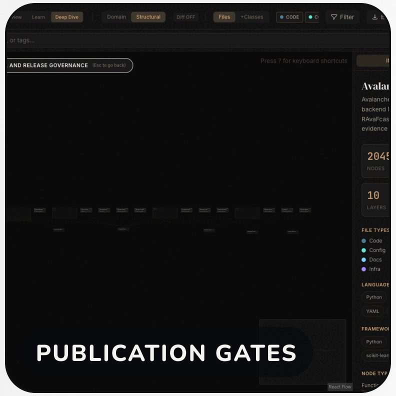
Many small votes

Each tree learns a different split pattern; the ensemble combines them to reduce dependence on a single brittle rule.

PUBLICATION GATE

Prediction is not enough

The score must still pass model-status, lineage, freshness, and confidence controls before it is safe to publish.



BACKEND/MODELS/SURROGATE_RF.PY • BACKEND/DAILY_INFERENCE.PY

Calibration And Explainability

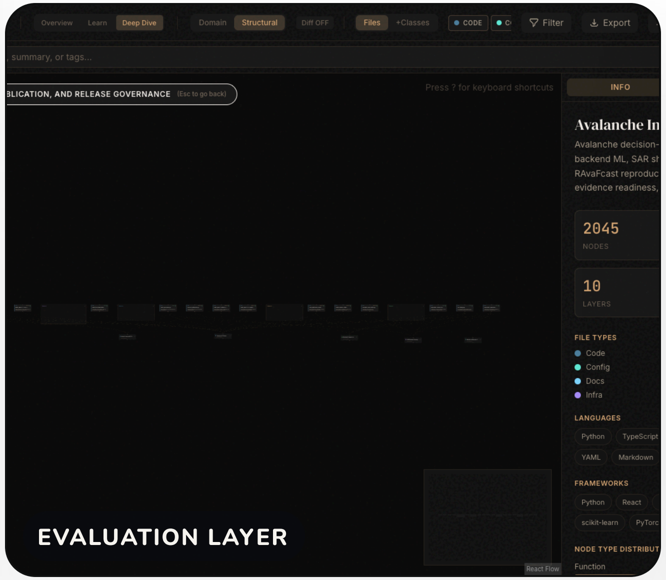
CALIBRATION

Calibration checks whether a probability behaves like a probability. If the model says 70% risk often, roughly 7 out of 10 comparable cases should behave that way over a valid evaluation set.

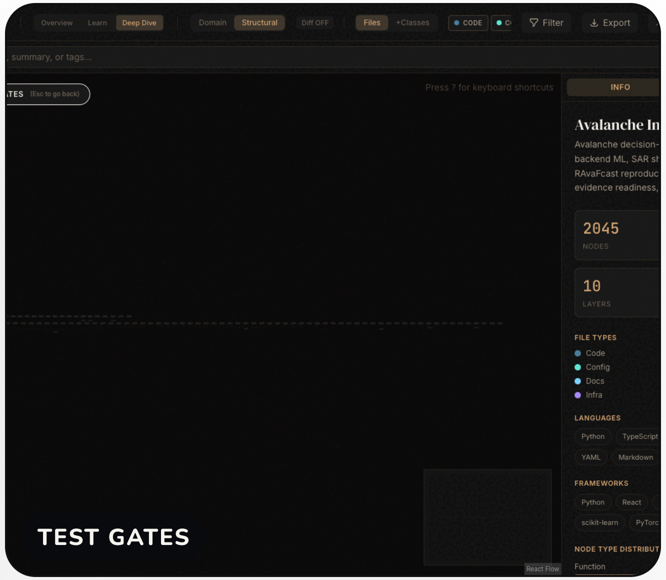
EXPLAINABILITY

TreeSHAP-style explanations and feature summaries help reviewers see why a cell moved up or down, instead of treating the model as a black box.

The current deck keeps explanation and calibration as evidence controls, not as magic. If a run falls back to heuristic explanation, that must remain visible.



EVALUATION LAYER



TEST GATES

CANDIDATE MODEL

Why MTS-LSTM May Help Later

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BEGINNER VERSION

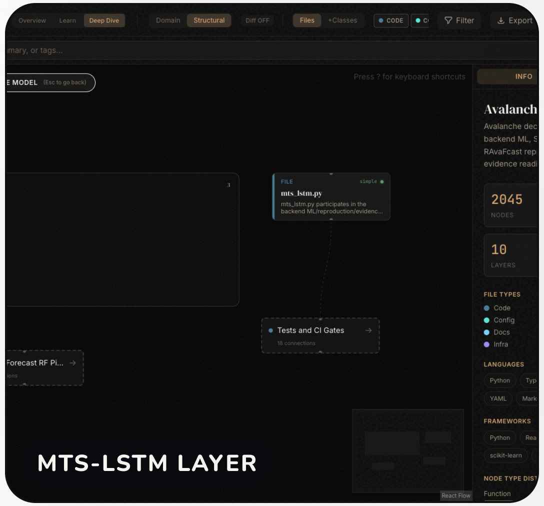
Random Forest sees a structured snapshot. MTS-LSTM can learn a short story over time: wind loading, snowfall, warming, settling, and changing instability.

Potential benefit

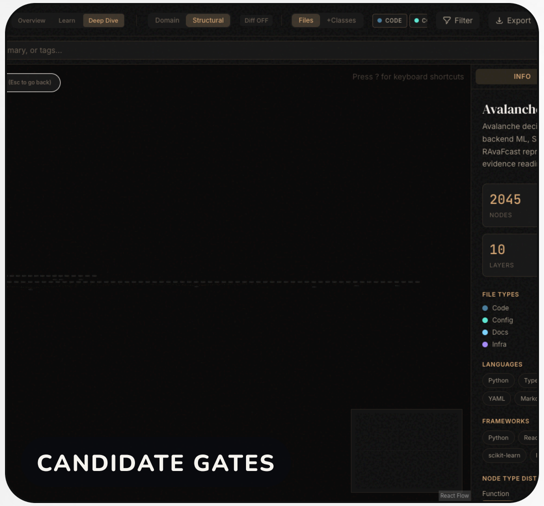
Better use of multi-day weather and snowpack sequences.

Current boundary

Candidate lane only; it needs reviewed data, benchmark gates, and drift checks before activation.



MTS-LSTM LAYER



CANDIDATE GATES

TECHNICAL EVIDENCE

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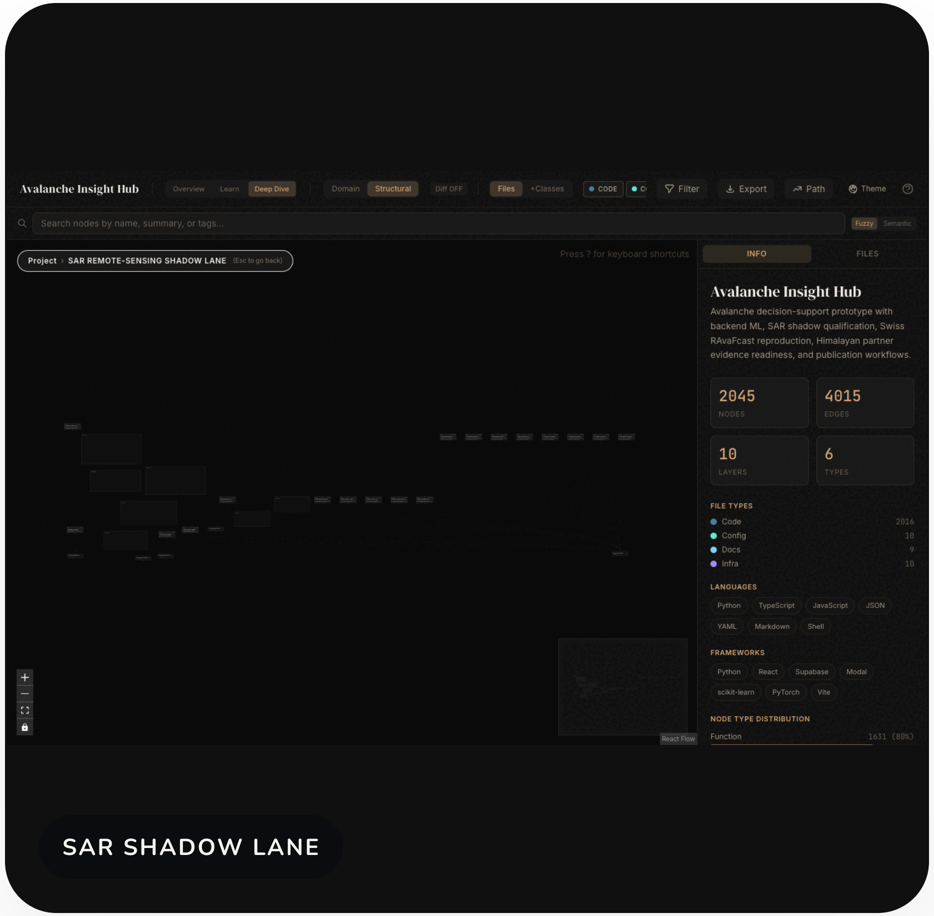
SAR Shadow Lane



SAR SHADOW LANE

Why it helps

Sentinel-1 can monitor large remote areas through cloud cover.

Why it is gated

Wet snow, crevasses, shadow/layover, small events, and scene transfer can distort detections.

Current posture

AvalCD and SnowSlide evidence are valuable, but SAR stays out of public scoring until held-out gates pass.

What Europe Adds To The Method

TECHNICAL EVIDENCE

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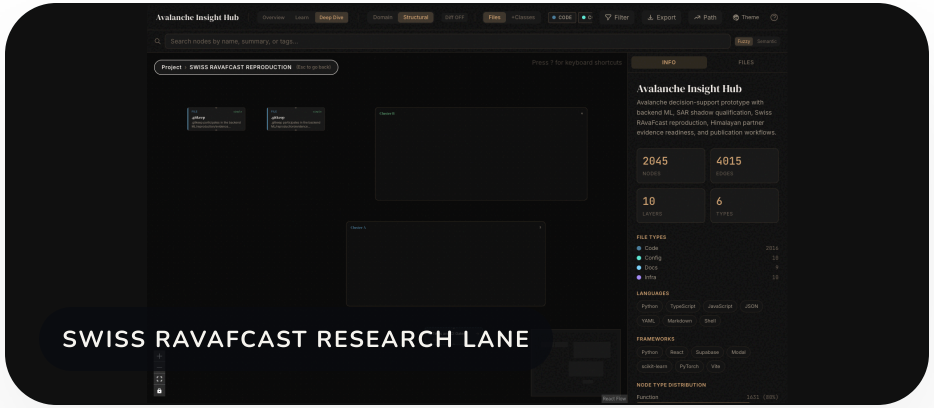

European method lessons

EAWS scale
Shared 1-5 danger language.

NHESS 2022
`D_tidy` labels and Random Forest baseline discipline.

GMD 2024
Swiss RAvaFcast RF, GPxyz, and refined aggregation.

The Cryosphere 2024
SAR transfer limits, wet-snow false positives, small misses, shadow/layover.

AvalCD
SAR change-detection benchmark context.

Claim discipline
Method transfer only; Himalayan proof needs local reviewed data.

EAWS • NHESS 2022 • GMD 2024 • THE CRYOSPHERE 2024 • AVALCD ARXIV 2026

Why This Helps Himalayas, But Is Not Enough

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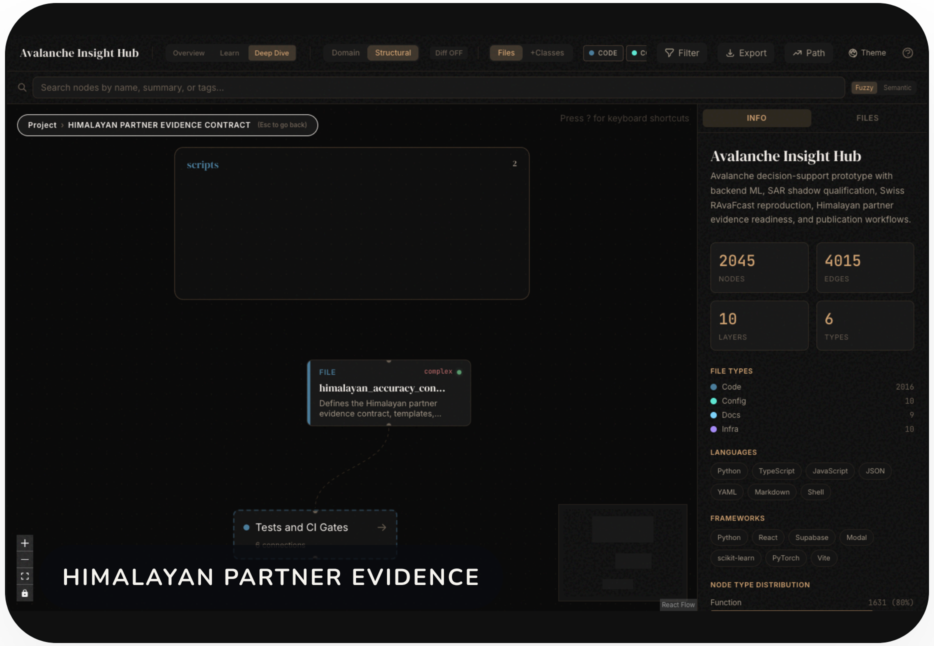


CRITICAL BOUNDARY

Swiss and European evidence can improve the playbook, feature discipline, and validation gates. It cannot substitute for Himalayan station X/Y/Z metadata, D_tidy-grade local labels, event evidence, polygons, and a leakage-checked holdout.

PARTNER CHECKLIST

Himalayan Data Checklist

RESEARCH AGENDA

Core files to fill

Station X/Y/Z

`station_metadata.csv`

Weather

`weather_station_observations.csv`

Snowpack

`snowpack_profile_features.csv`

D_tidy labels

`danger_labels_and_bulletins.csv`

Events and terrain

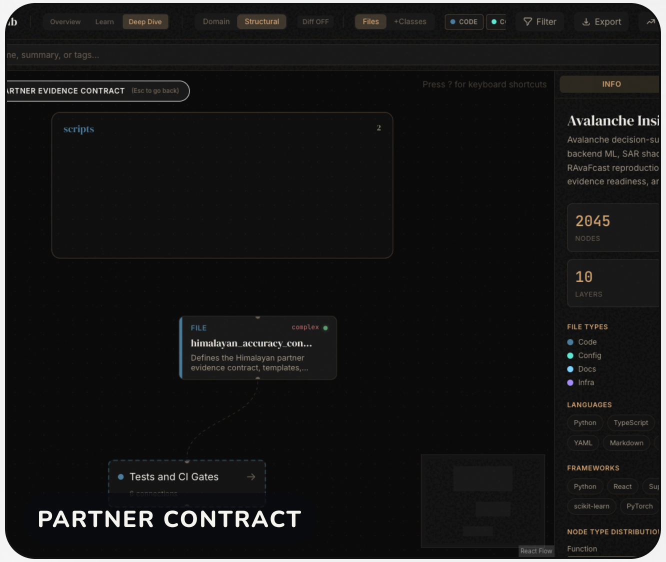
`historical_avalanche_events.csv`, `terrain_ates_runout_validation.csv`

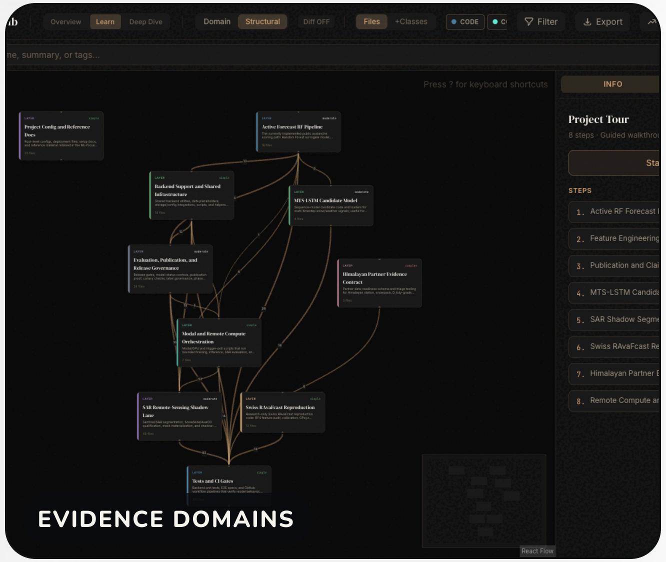
Regions and holdout

`warning_region_polygons.csv`, `independent_himalayan_holdout.csv`

FIELDS THAT MATTER MOST

- `source_ref`, `license_scope`, `review_status`, `reviewer_id`, and `reviewed_at`.
- Latitude, longitude, elevation, validity windows, avalanche regime, label source, and reviewer notes.





TECHNICAL EVIDENCE

Beginner FAQ

React + Vite

Builds the website screens and packages the frontend quickly.

Supabase + PostGIS

Stores database tables, auth, storage, functions, and geospatial map facts.

Python + scikit-learn

Runs data preparation, Random Forest, calibration, evaluation, and reports.

PyTorch

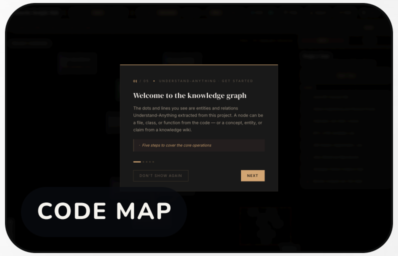
Supports MTS-LSTM candidate models and SAR segmentation experiments.

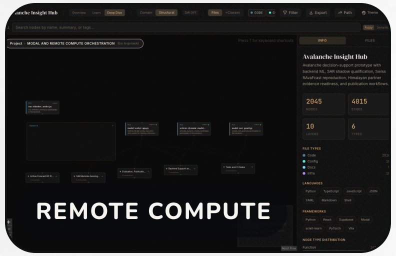
Modal

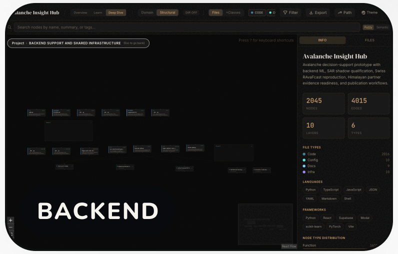
Runs bounded heavier compute jobs outside the public click path when authorized.

Understand Anything

Creates visual codebase maps so reviewers can see how the system fits together.







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RESEARCH AGENDA

What To Do Next, And What Not To Say

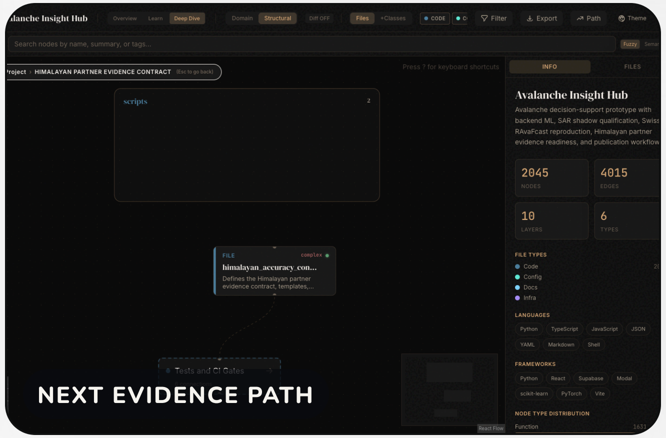
NEXT ACTIONS

- Send partner handoff packet and ask for reviewed Himalayan source packages.
- Run partner package triage and source-manifest validation.
- Have scientists adjudicate label quality, station coverage, event truth, and holdout validity.
- Advance only a narrow pilot if local gates pass.

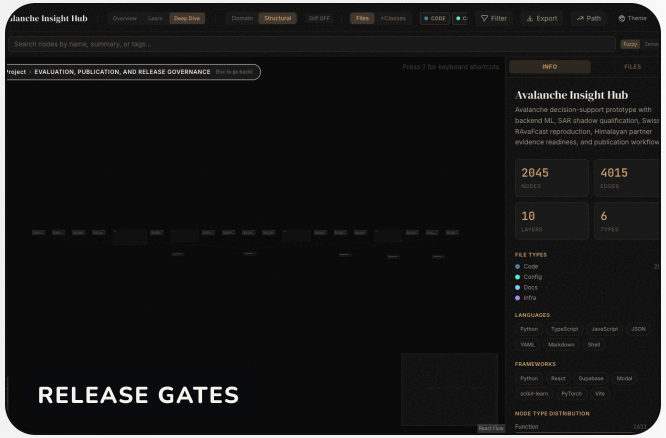
Avoid saying: Himalayan accuracy is established, SAR is activated for public scoring, or the platform replaces regional avalanche forecasters.

ALLOWED CURRENT FRAMING

Evidence-governed MVP V2 with a live technical proof region, active Random Forest baseline, research-only Swiss and SAR lanes, and a Himalayan partner evidence path.



NEXT EVIDENCE PATH



RELEASE GATES